

## ROST, Norway

WMO: 011070

Lat: 67.527N

Lon: 12.104E

Elev: 4

StdP: 101.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -4.7       | -3.5 | -10.5                         | 1.5 | -3.1 | -9.3 | 1.7 | -2.2 | 21.7                  | 1.9  | 19.5 | 2.6  | 6.8                   | 90   | 0.882 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.3                    | 17.1            | 14.3 | 15.7 | 13.5 | 14.7 | 12.8 | 14.7                | 16.5 | 13.8 | 15.4 | 13.1 | 14.4 | 4.7                  | 90   |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 13.7                            | 9.8 | 15.6 | 12.9 | 9.3 | 14.8 | 12.2 | 8.8 | 14.0 | 41.1          | 16.6 | 38.9 | 15.6 | 36.9 | 14.5 | 18.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 17.4              | 15.4 | 13.8 | DB                         | -5.2 | 19.0 | 1.7                | 2.4                                                | -6.4      | 20.7 | -7.4       | 22.1 | -8.3       | 23.4 | -9.5       | 25.2 |
|                   |      |      | WB                         | -6.7 | 15.6 | 1.6                | 1.5                                                | -7.8      | 16.7 | -8.8       | 17.6 | -9.7       | 18.4 | -10.8      | 19.5 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov  | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)  | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 6.0    | 1.1   | 1.4   | 1.6   | 4.0   | 6.7   | 8.9   | 11.8  | 12.0  | 10.4  | 7.0   | 4.3  | 2.5  |
|                                                                                  | DBStd       | 4.50   | 2.52  | 2.58  | 2.34  | 2.16  | 2.19  | 1.89  | 2.08  | 1.80  | 1.96  | 2.30  | 2.58 | 2.43 |
|                                                                                  | HDD10.0     | 1635   | 277   | 239   | 262   | 181   | 106   | 45    | 4     | 2     | 19    | 97    | 170  | 232  |
|                                                                                  | HDD18.3     | 4506   | 535   | 473   | 520   | 431   | 362   | 284   | 203   | 196   | 239   | 352   | 420  | 491  |
|                                                                                  | CDD10.0     | 171    | 0     | 0     | 0     | 0     | 3     | 11    | 59    | 66    | 29    | 3     | 0    | 0    |
|                                                                                  | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |
|                                                                                  | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |
| Wind                                                                             | WSAvg       | 7.0    | 8.1   | 8.0   | 7.9   | 6.6   | 6.2   | 6.3   | 5.6   | 5.5   | 6.1   | 7.6   | 7.5  | 8.2  |
| Precipitation                                                                    | PrecAvg     | 909    | 110   | 85    | 77    | 50    | 39    | 42    | 54    | 67    | 81    | 96    | 100  | 107  |
|                                                                                  | PrecMax     | 1262   | 192   | 155   | 133   | 100   | 64    | 67    | 85    | 108   | 122   | 183   | 199  | 208  |
|                                                                                  | PrecMin     | 536    | 50    | 27    | 12    | 24    | 17    | 16    | 22    | 28    | 30    | 26    | 17   | 27   |
|                                                                                  | PrecStd     | 165    | 37    | 35    | 32    | 18    | 11    | 15    | 17    | 21    | 22    | 43    | 39   | 44   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 7.3   | 7.1   | 7.1   | 10.1  | 14.7  | 16.0  | 19.9  | 18.6  | 15.6  | 13.1  | 10.6 | 8.1  |
|                                                                                  |             | MCWB   | 6.3   | 6.1   | 5.7   | 7.1   | 11.0  | 12.9  | 16.0  | 15.5  | 12.9  | 10.3  | 9.6  | 7.2  |
|                                                                                  | 2%          | DB     | 6.5   | 6.5   | 6.2   | 8.7   | 12.2  | 13.6  | 17.9  | 16.6  | 14.5  | 12.0  | 9.7  | 7.2  |
|                                                                                  |             | MCWB   | 5.4   | 5.5   | 5.0   | 6.4   | 9.5   | 11.2  | 14.8  | 14.2  | 12.4  | 9.9   | 8.3  | 6.1  |
|                                                                                  | 5%          | DB     | 5.7   | 5.9   | 5.6   | 7.6   | 10.6  | 12.4  | 16.3  | 15.5  | 13.7  | 11.1  | 8.7  | 6.5  |
|                                                                                  |             | MCWB   | 4.6   | 4.8   | 4.4   | 5.9   | 8.4   | 10.4  | 14.0  | 13.5  | 11.9  | 9.3   | 7.2  | 5.1  |
|                                                                                  | 10%         | DB     | 4.8   | 5.2   | 5.0   | 6.9   | 9.4   | 11.5  | 14.9  | 14.7  | 13.0  | 10.2  | 7.8  | 5.8  |
|                                                                                  |             | MCWB   | 3.5   | 4.1   | 3.7   | 5.4   | 7.6   | 9.8   | 13.0  | 13.0  | 11.5  | 8.6   | 6.3  | 4.2  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 6.6   | 6.5   | 6.1   | 7.5   | 11.2  | 13.0  | 16.6  | 15.8  | 13.7  | 11.1  | 9.9  | 7.5  |
|                                                                                  |             | MCDB   | 7.1   | 6.8   | 6.7   | 9.3   | 14.2  | 15.3  | 18.9  | 17.9  | 14.8  | 12.4  | 10.8 | 7.7  |
|                                                                                  | 2%          | WB     | 5.6   | 5.7   | 5.3   | 6.8   | 9.7   | 11.6  | 15.0  | 14.6  | 12.7  | 10.2  | 8.6  | 6.5  |
|                                                                                  |             | MCDB   | 6.3   | 6.3   | 6.0   | 8.0   | 11.8  | 13.1  | 17.2  | 16.1  | 13.8  | 11.5  | 9.6  | 7.0  |
|                                                                                  | 5%          | WB     | 4.7   | 5.0   | 4.7   | 6.3   | 8.7   | 10.7  | 14.1  | 13.8  | 12.2  | 9.6   | 7.4  | 5.4  |
|                                                                                  |             | MCDB   | 5.6   | 5.7   | 5.5   | 7.3   | 10.3  | 12.0  | 16.1  | 15.2  | 13.3  | 10.9  | 8.5  | 6.2  |
|                                                                                  | 10%         | WB     | 3.7   | 4.2   | 3.9   | 5.7   | 7.9   | 10.0  | 13.2  | 13.2  | 11.6  | 8.8   | 6.4  | 4.4  |
|                                                                                  |             | MCDB   | 4.8   | 5.1   | 4.9   | 6.7   | 9.2   | 11.3  | 14.7  | 14.4  | 12.8  | 10.1  | 7.6  | 5.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 2.9   | 2.8   | 3.0   | 3.1   | 3.3   | 3.0   | 3.4   | 3.3   | 2.9   | 2.6   | 2.7  | 2.8  |
|                                                                                  |             | MCDBR  | 3.4   | 3.0   | 3.0   | 3.6   | 5.0   | 4.5   | 5.8   | 4.4   | 3.5   | 2.7   | 2.6  | 3.4  |
|                                                                                  |             | MCWBR  | 3.7   | 3.2   | 3.1   | 2.6   | 3.1   | 2.9   | 3.5   | 2.9   | 2.4   | 2.4   | 2.9  | 3.7  |
|                                                                                  | 5% WB       | MCDBR  | 3.4   | 3.0   | 2.9   | 3.3   | 4.6   | 4.2   | 5.4   | 4.1   | 3.3   | 2.5   | 2.6  | 3.7  |
|                                                                                  |             | MCWBR  | 3.8   | 3.2   | 3.1   | 2.7   | 3.1   | 3.0   | 3.5   | 3.0   | 2.6   | 2.5   | 3.1  | 4.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.372  | 0.314 | 0.337 | 0.351 | 0.351 | 0.348 | 0.353 | 0.346 | 0.334 | 0.309 | 0.354 | N/A  |      |
|                                                                                  | taud        | 2.437  | 2.157 | 2.281 | 2.376 | 2.452 | 2.502 | 2.509 | 2.521 | 2.498 | 2.256 | 2.433 | N/A  |      |
|                                                                                  | Ebn at Noon | 110    | 478   | 697   | 802   | 846   | 859   | 840   | 808   | 719   | 483   | 102   | 0    |      |
|                                                                                  | Edh at Noon | 15     | 50    | 76    | 90    | 92    | 90    | 87    | 78    | 62    | 44    | 13    | 0    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.03   | 0.34  | 1.42  | 3.16  | 4.74  | 5.27  | 4.68  | 3.48  | 1.89  | 0.63  | 0.09  | 0.00 |      |
|                                                                                  | RadStd      | 0.00   | 0.04  | 0.15  | 0.29  | 0.33  | 0.70  | 0.44  | 0.29  | 0.16  | 0.11  | 0.01  | 0.00 |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (15 neighbors) | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -123    | N/A     | N/A     |

Nomenclature: See separate page